

B.Tech III Year I Semester (R20) Regular & Supplementary Examinations November/December 2024

MICROPROCESSORS AND MICROCONTROLLERS

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) What is the difference between the microprocessor and microcontroller? | 2M |
| (b) What do you mean by pipelining? | 2M |
| (c) List the different types of addressing modes of 8086. | 2M |
| (d) State and explain the different instruction formats of 8086. | 2M |
| (e) What is mode 0 operation of Programmable Peripheral Interface 8255? | 2M |
| (f) What are the modes of operation supported by 8255? | 2M |
| (g) Give the addressing modes of 8051. | 2M |
| (h) What is SFR? | 2M |
| (i) What is the timer registers? | 2M |
| (j) Differentiate the mode 0 and mode 1 operation of the timer. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Draw and explain the architecture of 8086 in detail. 5M
- (b) If code segment register is 2345H and instruction pointer is 1234H. Then find the physical address of given instruction by using with and without formula. 5M

OR

- 3 (a) What is Memory Banking? Explain in detail? 5M
- (b) Draw the 16-bit flag register of 8086 and List the flags and status of each bit. 5M
- 4 (a) List the classify the various group of Instruction and explain the Data transfer instruction in detail. 5M
- (b) Write a program to add a data byte located at offset 0500H in 2000H segment to another data byte available at 0600H in the same segment and store the result at 0700H in the same segment. 5M

OR

- 5 (a) Explain the Unconditional and conditional control transfer or branching instructions in detail. 5M
- (b) Add the contents of the memory location 2000H:0500H to contents of 3000H:0600H and store the result in 5000H:0700H. 5M
- 6 Draw the architecture and pin diagram of 8255 programmable peripheral interface (PPI) and explain the function of each block in detail. 10M

OR

- 7 With the help of neat diagram explain how 8251 USART is interfaced with 8086 and used for serial Communication? 10M
- 8 Draw the Pin Diagram of 8051 and explain the function of various signals. 10M

OR

- 9 List the various Instructions available in 8051 microcontroller and explain. 10M
- 10 Describe the different modes of operation of timers/counters in 8051 with its associated register. 10M

OR

- 11 With a neat circuit diagram explain how a 4x4 keypad is interfaced with 8051 microcontroller? 10M
